

Ilyass Ouardi

M.Sc. Computer Science, Università degli Studi di Milano
Mechanistic interpretability and AI Safety researcher
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Research Interests

I am interested in **AI safety** and **mechanistic interpretability**. To me, interpretability is not an academic exercise for its own sake, but a pragmatic tool to make model safety measurable and verifiable on the path toward safe AGI. Right now I am particularly focused on the **geometry of representations**: how abstract concepts — like refusal and truth — are encoded within a model's activation space. Two questions follow from this. The first is one of *model biology*: whether such concepts conform strictly to single linear directions, or rely instead on higher-dimensional, possibly non-linear manifolds. The second is about *applied interpretability*: whether representation-level edits — steering, abliteration, concept erasure — are genuinely surgical when evaluated against practical baselines, with off-target side-effects measured as rigorously as the primary behavioral change. The two are connected, since an intervention built on the wrong picture of the geometry is precisely one whose side-effects go unmeasured.

Education

M.Sc. in Computer Science, University of Milan, Italy — Sep 2024 – Present
Thesis: *What Does Forgetting Look Like? Where activations land after concept erasure*
Advisors: Prof. Alfio Ferrara & Elisabetta Rocchetti
Weighted average: **29.22/30** (3.90/4.00)

Selected coursework: Natural Language Processing (30 *cum laude*), Distributed and Pervasive Systems (30 *cum laude*), Security of Service-Oriented Architectures (30 *cum laude*), Advanced Biometrics (30), Audio Pattern Recognition (30), Enterprise Security Management (30), Privacy and Data Protection (29)

B.Sc. in Computer Science, University of Milan, Italy — Sep 2020 – Oct 2024
Thesis: *Design and development of a donation-tracking system for RECUP APS*
Advisor: Prof. Carlo Maria Bellettini
Final grade: **110/110 *cum laude***

Mathematical and theoretical foundations: Discrete Mathematics (30 *cum laude*), Mathematical Logic (30 *cum laude*), Automata and Formal Languages (30 *cum laude*), Programming Languages (30 *cum laude*), Artificial Intelligence I (30 *cum laude*), Calculus (30), Statistics and Data Analysis (30), Algorithms and Data Structures (30)

Diploma, Computer Science (Perito Informatico), "G. e M. Montani" Institute, Fermo — Sep 2015 – Jul 2020
Final grade: **100/100 *cum laude***

Research Experience

Independent Research — Representation Geometry of Truth, University of Milan — Dec 2025 – Jul 2026
Sole author. Natural Language Processing course.

- Tested whether propositional truth is mediated by one linear direction or by a k-dimensional concept cone. Adapted from Wollschläger et al. (2025), built **TDO** and **TCO**: gradient-optimised directions and cones under causal necessity, sufficiency and KL-retention losses. Evaluated on six instruction-tuned models from three families (Qwen-2.5 1.5/7/14B, Gemma-2 2/9B, Llama-3.1-8B).
- **Result:** a single direction is insufficient for the larger models, and a low-dimensional cone (k=2) is the better model of the concept. Its basis is near-orthogonal to the Difference-in-Means probe yet individually causally effective. For the three larger models, a single direction also exceeded the KL side-effect threshold on unrelated prompts; a 2D cone resolved both effectiveness and side effects for every model.
- **Manuscript:** *The Multidimensional Geometry of Truth in LLMs: A Concept-Cone Extension of the Linear Representation Hypothesis* (2026). [Draft] [Code]

Research Project — What a Heart-Sound Classifier Actually Learns, University of Milan Apr 2026 – Jul 2026
Sole author. Audio Pattern Recognition course.

- Compared three feature representations against three classifier families on PhysioNet/CinC 2016 under one recording-level protocol, pairing every performance claim with a diagnostic capable of refuting it.
- **Result:** the headline MAcc of 0.866 is largely a recording-site artefact — site recoverable from the same features at 0.647 against 0.167 chance, diagnosis collapsing from 0.840 to 0.457 under leave-one-site-out. Attribution leans systolic, but no more so on correct than on incorrect predictions.
- **Manuscript:** *What Does a Heart Sound Classifier Actually Learn?* (2026). [Draft] [Code]

Research Project — Empirical Test of the Legends Hypothesis, SESAR Lab, University of Milan Feb 2026 – Jun 2026
With F. Bylyshi & H. El-Khazri (equal contribution)

- Tested whether a model learns a regulation better from stories showing it being followed than from the regulation itself. Built a pipeline producing training corpora matched on everything but content, and **GenderEqGLUE**, a five-task benchmark evaluated on documents held out from training.
- **Result:** the two regimes teach complementary competences — training on the rule wins overall, training on the stories wins on the tasks that require applying it.
- **Manuscript:** *The Legend Challenge: Embedding Ethical Compliance into LLMs through Champion Narratives* (2026). [Draft] [Code]

Teaching Experience

Teaching Assistant, *Computer Programming (Python)* Oct 2026 – Present
University of Milan, Lecturer: Prof. Stefano Ferrari

- Teaching assistant for "Computer Programming", the first-year Python course of the inter-university Bachelor's in Artificial Intelligence (University of Pavia, University of Milan, University of Milano-Bicocca), taught in English.
- Support students during weekly lab sessions on imperative and object-oriented programming in Python; run weekly tutoring hours; assist with computer-based exams.

Teaching Assistant, *Accertamento delle Competenze Informatiche* (3 CFU) Jan 2026 – Sep 2026
University of Milan, Scientific lead: Prof. Stefano Montanelli; delivered by the CTU

- Prepared and delivered lectures to classes of students, authoring the accompanying teaching material.
- Led lab sessions of a blended-learning course shared across numerous undergraduate degree programmes, and supervised examinations.

Engineering Experience

Lead Developer / Consultant, RECUP APS – University of Milan Jan 2024 – Dec 2025

- Designed and built a donation-tracking web platform (React + TypeScript, Supabase DB/Auth/Storage, Netlify Functions); began as B.Sc. thesis, continued as a consultancy with me as lead developer.
- **RecupGPT** (2025): retrieval-augmented natural-language querying over the production database (LangChain, LangGraph, NL2SQL with query rewriting and routing).

Full-Stack Developer Consultant, Value Process Srl, Milan Jun 2022 – Sep 2022

- Microservices architecture in Spring Boot and React; led migration of a legacy system to a serverless architecture.

Fellowships, Awards, and Honours

AI Safety for Italy (AIS4I) — Italian AI safety community, Member and Organizer	Sep 2026 – Present
Lead the Future — STEM mentorship community, Mentee	Sep 2025 – Present
Full academic + merit scholarships (tuition and housing), University of Milan	Sep 2020 – Present
BOOST '24 — selected for the Bologna Orthogonal Summer Term (outstanding B.Sc./M.Sc. CS students)	Aug 2024
Futuro Annunciato '24 — selected for the event, for outstanding students	Jul 2024
Nomination, "Alfieri del Lavoro" Award, Presidency of the Italian Republic	Jul 2020
Participant, Olympiads in Informatics & Mathematics	2017 – 2020

Technical Skills

Mechanistic interpretability: denoising activation patching, directional ablation and activation addition, Difference-in-Means probes, gradient-based direction/cone optimisation, concept erasure (INLP, LEACE), activation steering

Post-hoc XAI: SHAP (tree and kernel estimators), Grad-CAM with saliency sanity checks, structured occlusion

Experimental method: causal mediation protocols, permutation and analytic nulls, grouped / leave-one-group-out cross-validation, paired significance testing with multiple-comparison correction, bootstrap over the correct unit of independence

ML & LLM tooling: PyTorch, nnsight, scikit-learn, LangChain, LangGraph, RAG, NL2SQL, supervised fine-tuning

Programming — proficient: Python, Java, TypeScript; *familiar*: Go, C, Scala, OCaml, Erlang

Web, data & cloud: React, SQL / PL-SQL, Supabase (PostgreSQL + RBAC), Netlify Functions, CI/CD

Languages: Italian (native), Arabic (native), English (fluent, working language)

Last updated: October 30, 2026